PORTFOLIO

Projects – Hailegiorgis Wagaye

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Summary

As a full-stack developer and software engineering student, I build secure, scalable systems using modern technologies. Experienced in cryptography, API design, cloud deployments, and enterprise-level applications. Passionate about tackling complex problems with innovative solutions and delivering production-ready software.

Key Strengths:

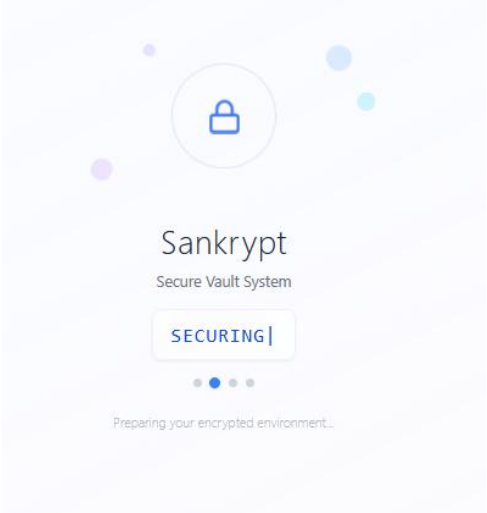
- Time Management – delivering high-quality software under tight deadlines
- Security Expertise – strong focus on secure system design and implementation
- Team Collaboration – excellent communication for diverse teams
- Fast Learning – quickly adapts to new technologies and complex problems

Key Achievements:

- Rebuilt and modernized the Aesop Meredaja platform (legacy PHP → Laravel + React + PostgreSQL)
- Developed a secure zero-knowledge password vault for client credentials
- Built multiple production-ready full-stack applications with documentation and cloud deployment

Sankrypt: Secure Zero-Knowledge Password Vault

Overview:
Sankrypt is a zero-knowledge password management platform that encrypts all sensitive data client-side before reaching the server. Designed for strong security, scalability, and modern web usage.



Architecture:

- Client-side encryption for all stored secrets
- Laravel API for secure backend services
- React frontend with modern UI
- Token-based authentication
- Encrypted storage in PostgreSQL

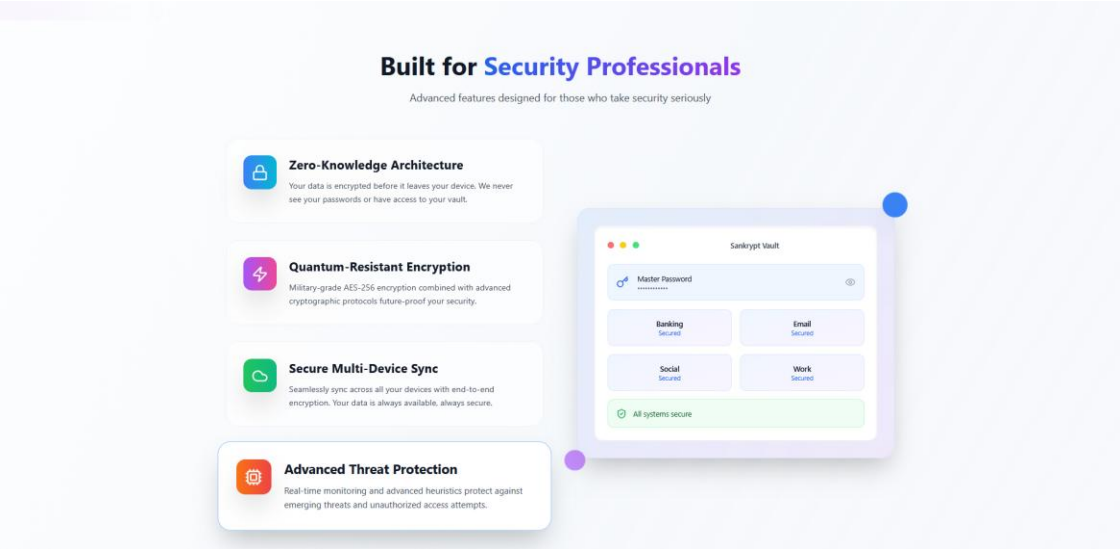
Key Features:

- Zero-knowledge password storage
- End-to-end encryption
- Secure vault management
- Browser extension support
- Production-ready deployment

Stack:
Laravel • React • PostgreSQL • Cryptography • REST API • Docker

Purpose:
Demonstrates secure system design, modern cryptographic practices, and full-stack

architecture for sensitive data applications.



See More: Live Demo

KD Poultry Farm Analytics System

Overview:
A data-driven farm management platform to track production, expenses, and profitability. Provides easy daily input tools and dashboards for non-technical users.

System Design:

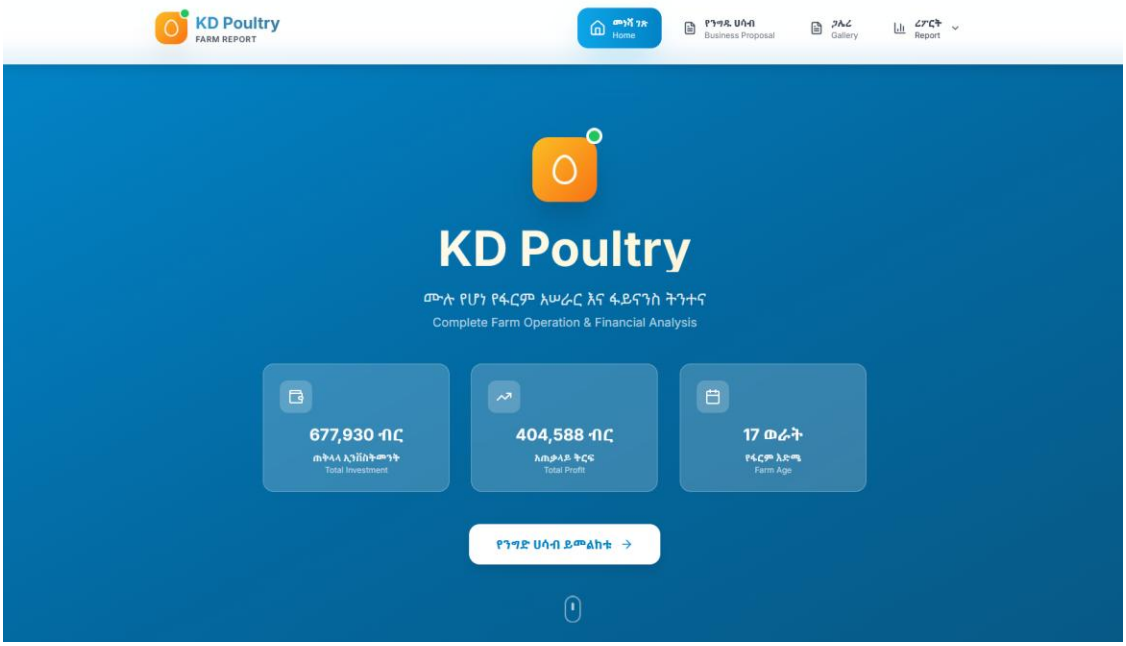
- React dashboard for farm operations
- Express backend for API and data processing
- Central database for production and financial tracking
- Visual reports with charts and summaries
- Optimized for low-resource environments

Key Features:

- Egg production and sales logging
- Expense and profit tracking
- Real-time analytics dashboard
- Simple UI for farm staff
- Scalable for multiple farms

Stack:
React • Node.js • Express • MySQL • Charts • REST API

Purpose:
Showcases practical software architecture and clear data visualization for business decision-making.

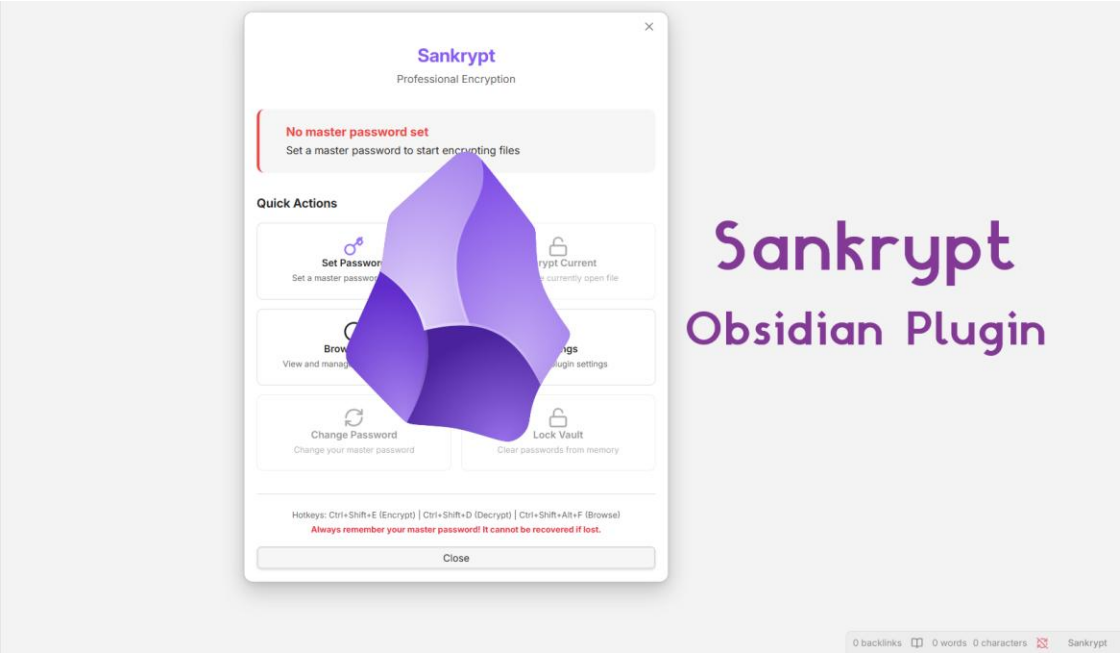


See More: Live Demo

Sankrypt: Professional File Encryption Plugin for Obsidian

Overview:
Professional AES-256-GCM encryption plugin for Obsidian to securely protect individual notes

and files. All operations are local with zero external dependencies.



System Design:

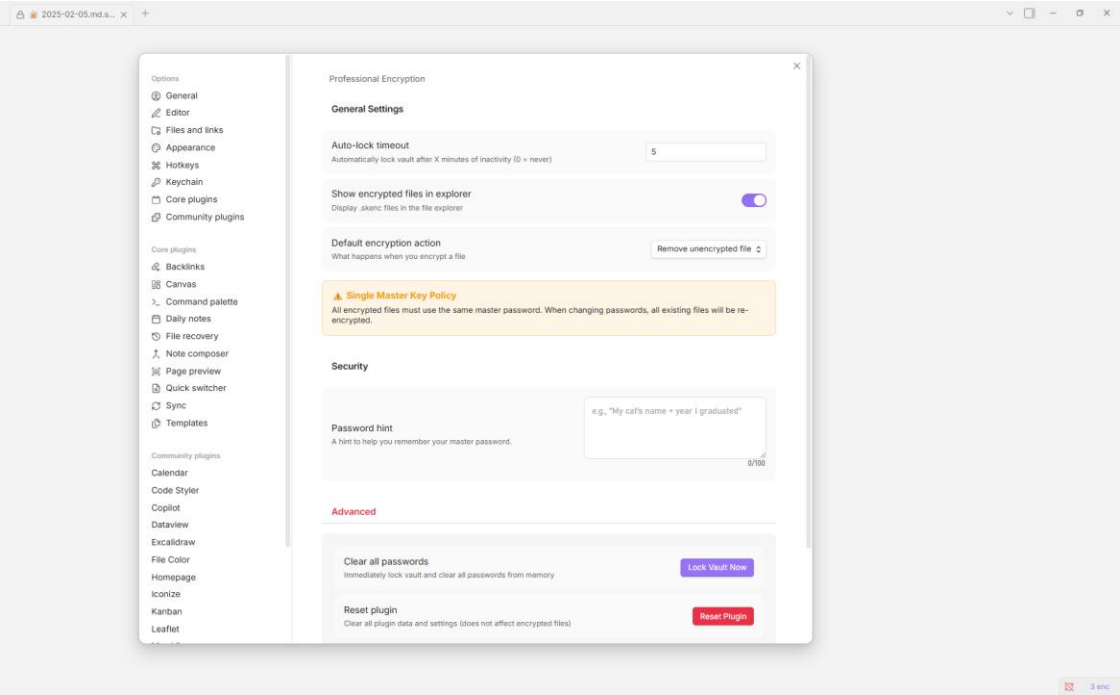
- Built with Obsidian Plugin API and modern JavaScript
- Uses Web Crypto API for zero-dependency encryption
- Single master password for all encrypted files
- Auto-lock feature clears passwords after inactivity
- Visual lock icons for encrypted files

Key Features:

- AES-256-GCM encryption for notes and files
- Smart filename conflict handling
- Secure session management
- Local-only operation, no internet access or telemetry
- Simple encrypt/decrypt workflow

Stack:
JavaScript • Obsidian Plugin API • Web Crypto API • AES-256-GCM

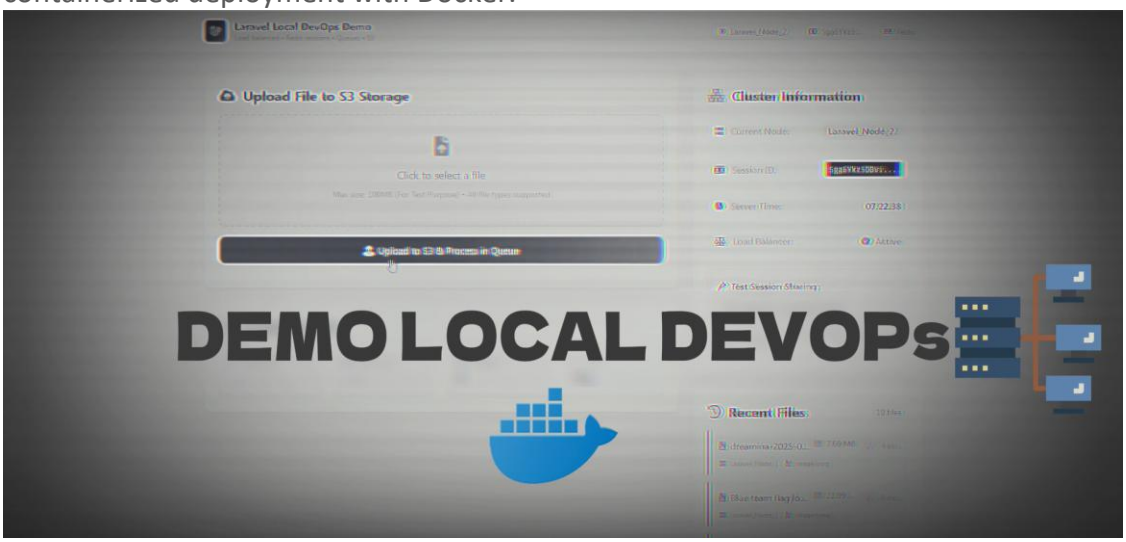
Purpose:
Provides secure client-side encryption for Obsidian users, protecting sensitive notes and personal data.



See More: GitHub

High-Availability Laravel Infrastructure with Docker & Load Balancing

Overview:
Production-style Laravel infrastructure demonstrating high availability, horizontal scaling, and containerized deployment with Docker.



Architecture:

- Nginx load balancing across multiple app servers
- Shared Redis sessions and cache
- Centralized MySQL database
- Queue workers for background jobs
- S3-compatible storage (MinIO → AWS/Hetzner ready)

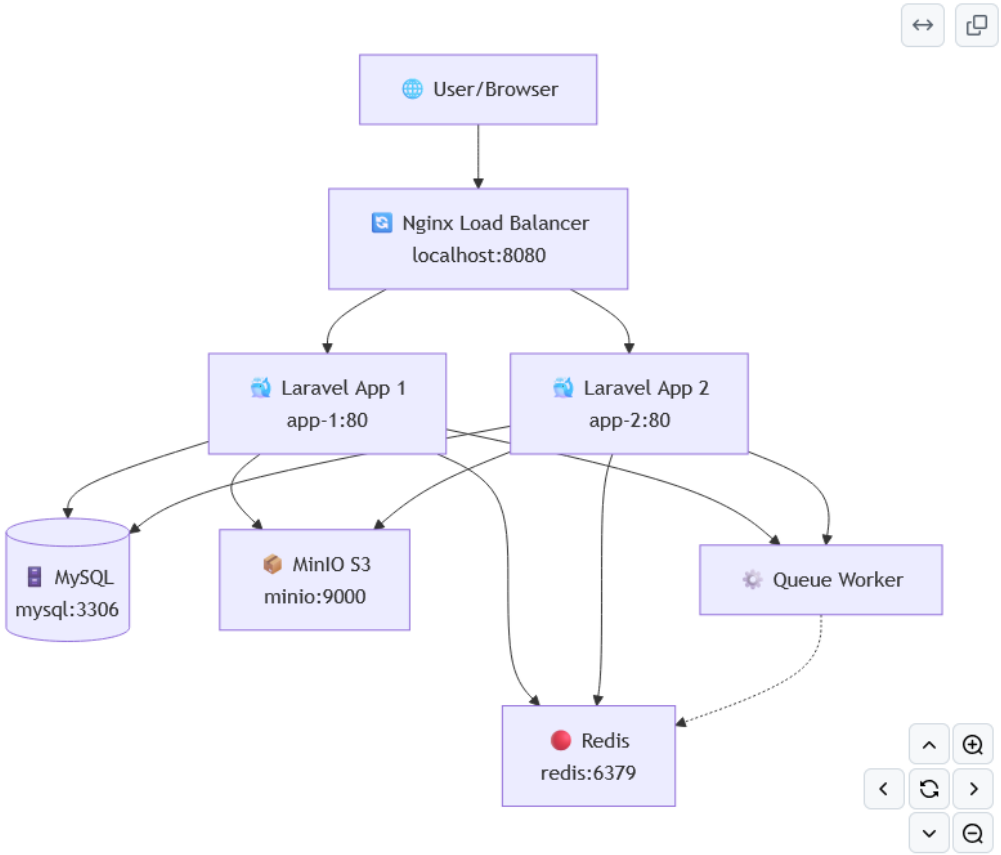
Key Features:

- Load balancing demonstration with two Laravel apps
- Session sharing with Redis
- Real-time queue processing
- Centralized file storage
- Production-ready container orchestration

Stack:
Docker • Nginx • Laravel • MySQL • Redis • MinIO • Queue Workers

Purpose:
Simulates real cloud deployments locally, enabling DevOps testing for high-availability Laravel

infrastructure.



See More: GitHub

Aesop Meredaja: Innovators & Investors Platform

Overview:
A full-featured platform connecting innovators with investors. Provides secure project sharing, encrypted messaging, and controlled investment workflows. Rebuilt from legacy PHP to a modern Laravel + React stack.

System Design:

- Laravel backend with secure APIs
- React frontend with responsive dashboards
- PostgreSQL database for structured data
- Payment gateway integration (PayPal, Chapa)
- TOTP-based authentication for enhanced security

Key Features:

- Secure project sharing & management
- End-to-end encrypted messaging
- Cap table simulation and investment tracking
- Multi-role dashboards (innovator, investor, admin)
- Production deployment with scaling and security optimizations

Stack:
Laravel • React • TypeScript • Tailwind CSS • PostgreSQL • REST API

Purpose:
Demonstrates enterprise-level full-stack development, secure workflow management, and scalable architecture for fintech/innovation platforms.

See More: Live Demo